Exercises

Algebraic problems

Question 4.1: Find the stationary point of the following functions and verify whether they are maxima or minima either by evaluating the function, or using the properties of the derivative.

(a) $f(x) = x^2 + x$
(b) $f(x) = 2x - x^2$
(c) $f(x) = (x-2)^2$
(d) $f(x) = (x-2)(3-x)$
(e) $f(x) = \frac{x^3}{3} - 4x^2 + 15x + 5$
(f) $f(x) = ln(x) - x$

Question 4.2: Find two non-negative numbers whose sum is 9 and so that the product of one number and the square of the other number is a maximum. To solve this, first construct the two equations that define these conditions, using variables x and y. To start, you can write $x + y = 9$.

Real-world problems

Question 4.3: Amazon wants to construct a box with length l, width w and height h which has volume:

$$V = l \times w \times h$$

and only has 10 m^2 of material to use in construction of the box:

$$10 = 2lw + 2wh + 2lh$$

If the box will have a square base (i.e. $l = w$) and assuming that all the material is used in the construction process, determine the maximum volume that the box can have. You will need to rearrange the constraints and substitute into the volume equation to find a function with respect to one variable before optimising.

Question 4.4: Amazon now wants to construct an open rectangular box with square base is to be made from 48 m^2 of material. What dimensions will result in a box with the largest possible volume?

Question 4.5: A cylindrical aluminium can of cola is to hold 330 ml of liquid. Coca-Cola would like to know what the dimensions of the can that uses the least amount of aluminium. The volume for a cylinder of base radius r and height h is given by

$$V = \pi r^2 h$$

The formula for surface area of a cylinder is given by:

$$A = 2\pi rh + 2\pi r^2$$

The volume is constrained so for whatever value of r we pick, we can calculate h and therefore estimate the surface area of the can. Rewrite the volume equation to get:

$$h = \frac{V}{\pi r^2} = \frac{330 \times 10^{-6}}{\pi r^2}$$

Substitute this into the surface area equation:

$$A = 2\pi rh + 2\pi r^2 = 2\pi r \times \frac{330 \times 10^{-6}}{\pi r^2} + 2\pi r^2 = \frac{660 \times 10^{-6}}{r} + 2\pi r^2$$

If you choose $r = 0.01$m, then, since the volume has to be 330×10^{-6}, the height, h, and surface area, A, has to be:

$$h = \frac{330 \times 10^{-6}}{\pi \times 0.01^2} = 1.050m \qquad A = \frac{660 \times 10^{-6}}{0.01} + 2\pi \times 0.01^2 = 0.067$$

(a) Complete the table below for surface area for different values of the radius r. Which is the best value to select?

Radius	Height for volume to equal 330 ml	Surface area of aluminium
0.010	1.050	0.067
0.020		
0.030		
0.040		
0.050		
0.060		
0.070		

(b) Use differentiation to find the minimum amount, what value do you arrive at?

Question 4.6: A gold seam optimisation problem. You decide to mine a gold seam and want to decide on how long to mine it for to maximise your profits. The cost of gold you can mine per day, G, is a function of the number of days, n, that you mine for. It becomes more difficult to mine the gold as time goes on:

$$G = 1 \times 10^5 - \frac{1 \times 10^5}{n+1}$$

In addition the cost of renting the equipment to mine the gold, C, increases with the number of days:

$$C = 1000 \times n$$

So that your profit is $P = G - C$. Graphs of P, G and C are shown in Figure 4.3.

(a) Work out how long you should mine the gold to optimise (find the stationary value of) your profit, P. (Hint: you can either use *chain or quotient rules* to perform the differentiation. To find the stationary points you will need to solve a quadratic equation).

(b) In the project a "snag" occurs that causes no profit to be made on one day (see Figure 4.4). After the snag is overcome mining can proceed at a faster rate. A function describing the cost of gold mined as the project goes on is:

$$G = 1 \times 10^5 \left[1 - \left(0.01n^2 + 1\right) \exp\left(-0.1n\right)\right]$$

Use this function to find the day where the amount of gold extracted hits a snag (you will need to make use of the product rule).

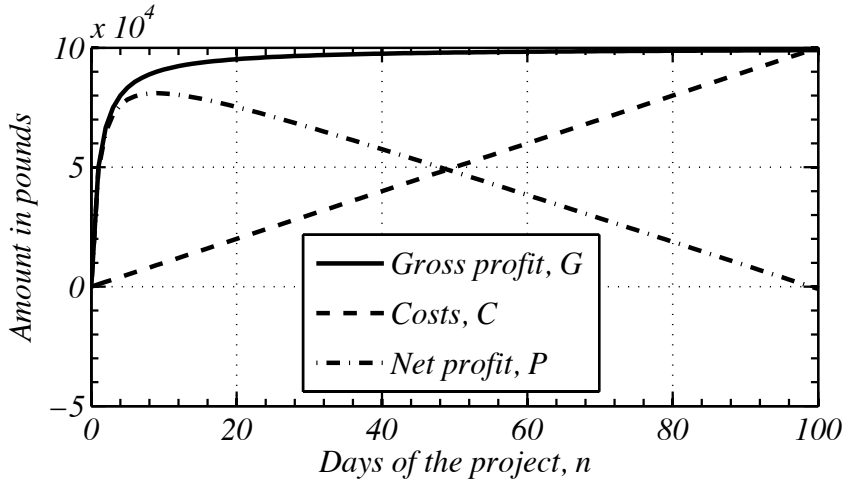


Figure 4.3: Hypothetical plot of profit as a function of time

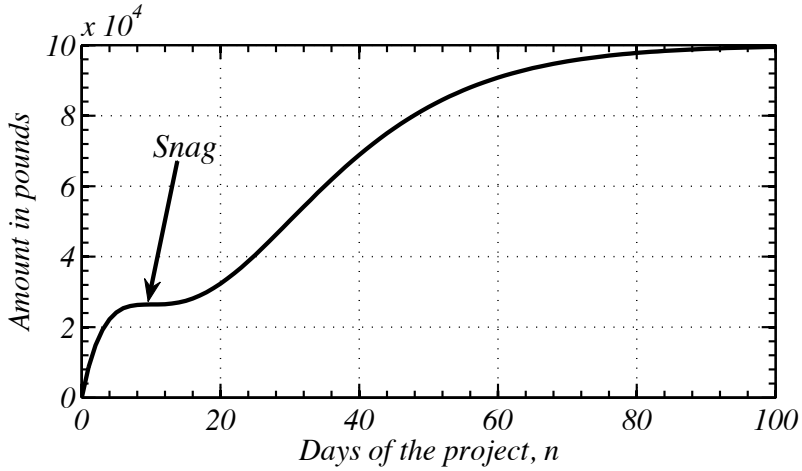


Figure 4.4: Hypothetical plot of profit as a function of time

Question 4.7: Geoengineering. In this problem economists investigating the *economic cost to society*, E, of an increase in global temperature, ΔT (e.g. see the Stern report) estimate that:

$$E = 500 \times \Delta T \tag{4.1}$$

with E in billions of dollars (see Figure 4.5). In addition climate scientists investigating the *marine cloud brightening* proposal for geoengineering estimate that the technique does have the capability to offset warming by rising CO_2. They estimate spending money, C, on a program to implement geoengineering can result in a lower increase in temperature:

$$\Delta T = 10 \exp(-dC) \tag{4.2}$$

where C is the amount spent on the scheme and d is a constant that has some uncertainty (see Figure 4.6).

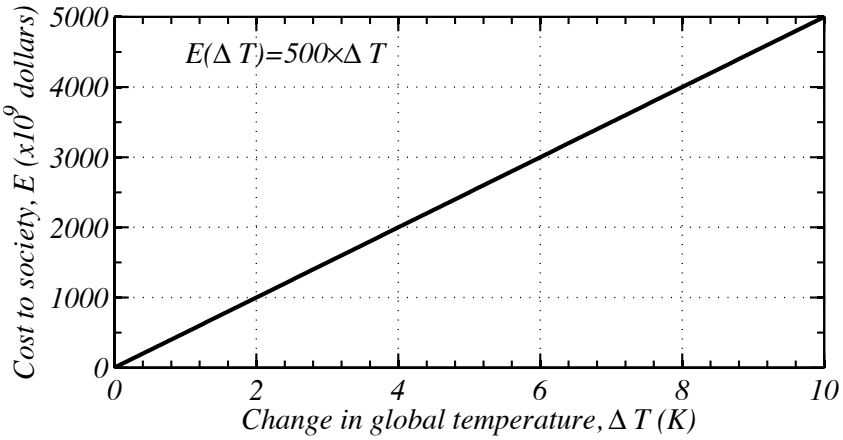


Figure 4.5: Hypothetical cost to society as a function of changes in global temperatures

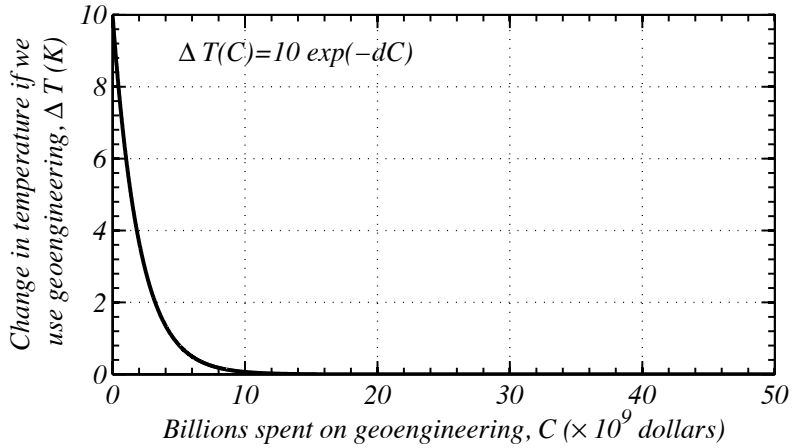


Figure 4.6: Hypothetical change in global temperatures as a function of investment in geo-engineering

In order to decide whether the scheme is worth implementing we need to get an idea of the *total cost* that implementing the scheme has (i.e. the benefit plus cost of scheme) vs the amount spent on implementing it. To do this rearrange Equation 4.1 to give $\Delta T = \frac{E}{500}$ and substitute into Equation 4.2 to get an equation for the economic cost to society, E, vs the amount spent on geoengineering, C. Then add on the cost of the scheme, C, to get the *total cost*, $T = E + C$:

$$T = 500 \times 10 \exp(-dC) + C \tag{4.3}$$

This function can be seen in Figure 4.7 and we can see there is a minimum in the *total cost*.

Your goal is to optimise / find the stationary value of Equation 4.3 (i.e. find how much we should spend on the scheme so that we can minimise the total cost). Choose the parameter d to be equal to 2×10^{-4}; how much should be spent? What is the meaning of your answer (i.e. what should the advice to policy makers be?)

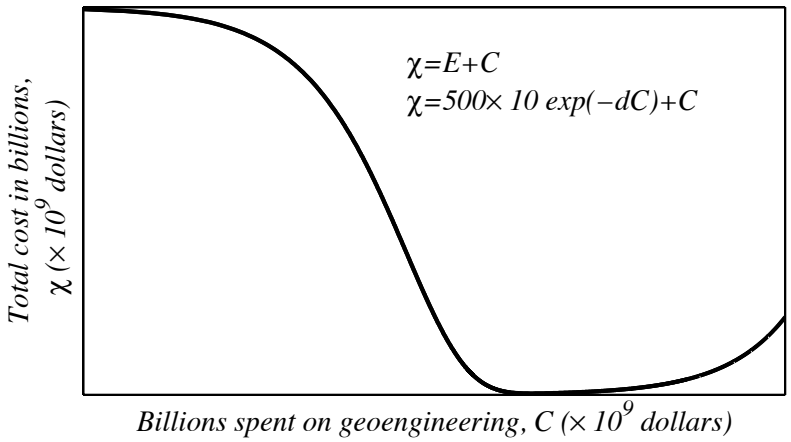


Figure 4.7: Hypothetical total cost to society as a function of investment in geo-engineering